

Assignment - 04

22BCON1068

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• Set-A ($2 \times 5 = 10$)Q1 Show that 1601 is a prime number.Sol Consider, the square root of 1601 which is approximately 40.012. If 1601 is not prime, it must have a division d such that $1 < d \leq 40$.Check if any prime number less than or equal to 40 divide 1601. Since, none of the divisors result in integer quotient, there is no division d such that $1 < d \leq 40$ for 1601.

Hence, we conclude that 1601 is a prime number.

Q2 Define inductive step of mathematical induction.Ans Inductive step: in inductive step, we show that the statement is true for some positive integer k , then it is also true for $k+1$. This means for all positive integer k , if $P(k)$ is true, then $P(k+1)$ is true.Q3 Prove that "for each integer n we have $\frac{3}{n^3-n}$ ".Sol To prove that "for each integer n we have $\frac{3}{n^3-n}$ ".We need to show that denominator n^3-n is never equal to zero for any integer.

$$n^3-n=0$$

$$n(n^2-1)=0$$

$$n(n+1)(n-1)=0$$

$$n=0, 1, -1$$

So, for each integer n except $n=0, 1, -1$ the expression $\frac{3}{n^3-n}$ is defined.

$$n^3-n$$

Q4. Prove that $\forall$ real numbers a and b , if $b^2 > a^2$ then $b > a$.

Sol. Proof by Contraposition:

Given Statement is of the form: "if $b^2 > a^2$, then $b > a$ "

The Contrapositive is: if $b \leq a$, then $b^2 \leq a^2$.

Lets, Prove the contrapositive.

Assume $b \leq a$, it means either $b < a$ or $b = a$.

- (i) if $b < a$, squaring both sides $b^2 < a^2$
- (ii) if $b = a$, squaring both sides $b^2 = a^2$

from both cases $b^2 \leq a^2$

Hence, we indirectly proven the given statement.

Q5. Show that $n < 2^n$, for every integer $n > 0$.

Sol. Let the given statement be $P(n)$

$$P(n) = 2^n > n$$

Base step: $P(1): 2^1 > 1$ which is true.

inductive step: $P(k): 2^k > k$ — (1)

Check Statement $P(n)$ for $n = k+1$

NOW, we shall prove that $P(k+1)$ is true whenever $P(k)$ is true multiply eqⁿ (1) by 2 both sides

$$2 \cdot 2^k > 2k$$

$$2^{k+1} > 2k$$

$$2^{k+1} > k+k$$

$$2^{k+1} > k+1$$

$$\therefore (k+k > k+1)$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true

Hence, by mathematical induction $P(n)$ is true for every positive integer n .

• PART - B

Q1. Show that $\sqrt{2}$ is irrational.Ans Suppose that $\sqrt{2}$ is a rational number. Then we can find integers a and b such that;

$$\sqrt{2} = \frac{a}{b}$$

and we can assume that any common factors in a and b have been cancelled. We now square both sides of equation to get

$$2 = \frac{a^2}{b^2}$$

Thus, $2b^2 = a^2$, Hence a^2 is a multiple of 2 and is therefore even. So a must be even, since the square of any odd number is also odd. Thus, 2 divides a and we can write $a = 2m$ for some integer m . Hence,

$$2b^2 = (2m)^2 \text{ or } 2b^2 = 4m^2 \text{ or } b^2 = 2m^2$$

Thus b^2 is even. Hence b is even. But now a and b have a common factor of 2, which is a contradiction to the statement that a and b have no common factors. Hence our initial assumption that $\sqrt{2}$ is a rational number is false. Thus,

~~$\sqrt{2}$ is a rational number.~~
 $\sqrt{2}$ is an irrational number.

Q2. Show that $10^{2n-1} + 1$ is divisible by 11 for each natural no. n .

(b) Prove by the principle of mathematical induction
 $1 \cdot 2 + 2 \cdot 2^2 + \dots + n \cdot 2^n = (n-1)2^{n+1} + 2, n \geq 1.$

Ans.

(1) We can write

 $P(n): 10^{2n-1} + 1$ is divisible by 11

Basic step

 $P(1): 10^{2 \cdot 1 - 1} + 1 = 10 + 1 = 11$ which is divisible by11. Thus $P(n)$ is true for $n=1$.

Inductive step:

Let $P(k)$ be true for some natural number k i.e., $P(k): 10^{2k-1} + 1$ is divisible by 11

We can write

$$10^{2k-1} + 1 = 11a \quad \text{--- (1) where } a \in \mathbb{N}$$

Now, we will prove that $P(k+1)$ is true whenever $P(k)$ is true.

$$10^{2(k+1)-1} + 1$$

$$= 10^{2k+1} + 1$$

$$= 10^2 (10^{2k-1}) + 1$$

$$= 10^2 (10^{2k-1} + 1 - 1) + 1 \quad (\text{added \& subtracted 1})$$

$$= 10^2 (11a - 1) + 1 \quad \text{from eqn. (1)}$$

$$= 10^2 \cdot 11a - 100 + 1$$

$$= 10^2 \cdot 11a - 99$$

$$= 11(100a - 99), \text{ which is divisible by 11.}$$

Thus $P(k+1)$ is true, whenever $P(k)$ is true.Hence, from the principle of mathematical induction, the statement $P(n)$ is true for all natural numbers i.e., $n \in \mathbb{N}$.

$$(b) \quad P(n): 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + n \cdot 2^n = (n-1)2^{n+1} + 2$$

Base step: for $n=1$

$$L.H.S = 1 \cdot 2$$

$$= 2$$

$$R.H.S = (1-1)2^{1+1} + 2 = 0 + 2$$

$$= 2$$

$$\therefore L.H.S = R.H.S$$

$\therefore P(1)$ is true.

inductive step: let us assume $P(k)$ is true for some $k \in \mathbb{N}$.

$$P(k): 1 \cdot 2 + 2 \cdot 2^2 + \dots + k \cdot 2^k = (k-1)2^{k+1} + 2 \quad \text{--- (1)}$$

$$P(k+1): 1 \cdot 2 + 2 \cdot 2^2 + \dots + k \cdot 2^k + (k+1) \cdot 2^{k+1} =$$

$$(k+1-1)2^{k+1+1} + 2$$

$$1 \cdot 2 + 2 \cdot 2^2 + \dots + k \cdot 2^k + (k+1) \cdot 2^{k+1} = k \cdot 2^{k+2} + 2 \quad \text{--- (2)}$$

We have to prove $P(k+1)$ from $P(k)$ i.e (2) from (1).

$$1 \cdot 2 + 2 \cdot 2^2 + \dots + k \cdot 2^k = (k-1)2^{k+1} + 2$$

Adding $(k+1) \cdot 2^{k+1}$ to both sides:

$$1 \cdot 2 + 2 \cdot 2^2 + \dots + k \cdot 2^k + (k+1) \cdot 2^{k+1} = (k-1)2^{k+1} + 2 + (k+1)2^{k+1}$$

$$= k \cdot 2^{k+1} - 2^{k+1} + 2 + k \cdot 2^{k+1} + 2^{k+1}$$

$$= 1 \cdot 2 + 2 \cdot 2^2 + \dots + k \cdot 2^k + (k+1) \cdot 2^{k+1} = 2k \cdot 2^{k+1} + 2$$

$$= k \cdot 2^{k+2} + 2$$

which is same as $P(k+1)$ or eqn (2)

$\therefore P(k+1)$ is true whenever $P(k)$ is true.

By the principle of mathematical induction,

$P(n)$ is true for $n, n \in \mathbb{N}$.

Q3 By the P. & M. induction, show that $3^{4n+2} + 5^{2n+1}$ is a multiple of 14, for all positive integer value of n including zero.

Ans

Using principle of mathematical induction:

Base case: $P(n): 3^{4n+2} + 5^{2n+1}$

$P(0): 3^2 + 5^1 = 14$ which is multiple of 14.

Base case is true.

Inductive step:

Assume statement is true for any positive integer k .

$P(k) = 3^{4k+2} + 5^{2k+1} = 14M$ is multiple of 14 — (1)

Base case is true.

Now, we have to prove that the statement is true for $P(k+1)$:

$$= 3^{4(k+1)+2} + 5^{2(k+1)+1}$$

$$= 3^{4(k+1)} \cdot 3^2 + 5^{2(k+1)} \cdot 5$$

$$= 3^{4(k+1)+2} + 5^{2(k+1)+1}$$

$$= 3^{4k+6} + 5^{2k+3}$$

$$= 3^{4k+2} \cdot 3^4 + 5^{2k+1} \cdot 5^2$$

$$= 3^4 \cdot 3^{4k+2} + 5^2 \cdot 5^{2k+1} \quad \text{--- (2)}$$

By inductive hypothesis, $3^{4k+2} + 5^{2k+1}$ is a multiple of 14, Therefore, we can write $3^{4k+2} + 5^{2k+1} = 14m$ for some integer m .

Substitute (1) into (2)

$$= 3^4 \cdot 14m + 5^2 \cdot 14m$$

$$= 14(81m + 25m)$$

This show that $3^{4(k+1)+2} + 5^{2(k+1)+1}$ is a multiple of 14. By mathematical induction, the statement is true for all positive integer, including zero.

• Part - C

Q1. (a) show that for any positive integer $n > 1$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

solⁿ let $P(n) : \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$

Base step: $P(2)$ is true since $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} = 1 + \frac{1}{\sqrt{2}} > \sqrt{2}$

inductive Step: let $P(k)$ is true, i.e., $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} > \sqrt{k}$

Then we need to show that $P(k+1)$ is true i.e;

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k+1}} > \sqrt{k+1}$$

L.H.S : $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}}$ $\therefore$ using $P(k)$

$$= \sqrt{k} + \frac{1}{\sqrt{k+1}}$$

$$= \frac{\sqrt{k}\sqrt{k+1} + 1}{\sqrt{k+1}}$$

$$= \frac{\sqrt{k}\sqrt{k+1} + 1}{\sqrt{k+1}} \times \frac{\sqrt{k+1}}{\sqrt{k+1}}$$

$$= \frac{\sqrt{k+1}(\sqrt{k}\sqrt{k+1} + 1)}{k+1} \quad (\because \sqrt{k+1} > \sqrt{k})$$

$$\geq \frac{\sqrt{k+1}(\sqrt{k}\sqrt{k} + 1)}{k+1}$$

$$= \frac{\sqrt{k+1}(k+1)}{(k+1)} = \sqrt{k+1} = \text{R.H.S of } P(k+1)$$

Thus $P(k+1)$ is true, Hence by mathematical induction $P(n)$ holds for every integer $n \geq 1$.

(b) Prove that for all integers $n \geq 4$, $3^n > n^3$
 Sol Let $P(n) = 3^n > n^3$, where $n \geq 4$

Base step: $P(n)$ is true since $3^4 = 81 > 4^3 = 64$

Inductive step: Let $P(k)$ is true for all $k \geq 4$, that is $3^k > k^3$. Then we need to show that $P(k+1)$ is true, i.e., $3^{k+1} > (k+1)^3$

$$\begin{aligned} \text{Let us rewrite } (k+1)^3 &= k^3 + 3k^2 + 3k + 1 \\ &= k^3 \left(1 + \frac{3}{k} + \frac{3}{k^2} + \frac{1}{k^3} \right) \end{aligned}$$

Since $3^k > k^3$ (using $P(k)$), we would be done if we could also prove that $3 > 1 + \frac{3}{k} + \frac{3}{k^2} + \frac{1}{k^3}$ for $k \geq 4$

Observe that function $f(k) = 1 + \frac{3}{k} + \frac{3}{k^2} + \frac{1}{k^3}$ decreases as k

increases, so that $f(k)$ is largest when k is smallest.

In other words, $f(4)$ is the largest value of $f(k)$, where $k \geq 4$. Since

$$f(4) = 1 + \frac{3}{4} + \frac{3}{16} + \frac{1}{64} = \frac{125}{64}$$

is obviously less than 3, we have for any integer $k \geq 4$, $3 > 1 + \frac{3}{k} + \frac{3}{k^2} + \frac{1}{k^3}$.

Thus combining the two facts:

$$3 > 1 + \frac{3}{k} + \frac{3}{k^2} + \frac{1}{k^3} \text{ and } 3^k > k^3 \text{ for } k \geq 4, \text{ we can}$$

multiply and get,

$$3^{k+1} > k^3 \left(1 + \frac{3}{k} + \frac{3}{k^2} + \frac{1}{k^3}\right)$$

$$\text{or } 3^{k+1} > (k+1)^3$$

So $P(k+1)$ is true and then by mathematical induction $P(n)$ is true for all integers $n \geq 4$ i.e., $3^n > n^3$.

Q2. Prove that for all integers, $3^n > n^3$.

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Using mathematical induction,

Base step: for $n=1$, $P(n) : 3^n > n^3$

$$P(1) : 3^1 > 1^3$$

$$3 > 1$$

$P(n)$ is true.

Inductive step:

Assume the inequality holds for some arbitrary positive integer k , i.e., assume $3^k > k^3$.

Now,

We want to prove that the inequality holds for $k+1$.

Consider $n = k+1$

$$3^{k+1} = 3^k \times 3$$

Using assumption $3^k > k^3$
 $\checkmark$ ~~3^k~~ $3^{k+1} > 3 \times k^3$

Now we need to show that

$$3 \times k^3 > (k+1)^3$$

$$3 \times k^3 > k^3 + 3k^2 + 3k + 1$$

$$3k^3 > k^3, \quad (\text{Comparing each term with } 3k^3)$$

$$3k^3 > 3k^2,$$

$$3k^3 > 3k,$$

$$3k > 1$$

Summing these inequalities, we get

$$3 \cdot k^3 > k^3 + 3k^2 + 3k + 1$$

So, $P(k+1)$ is true, then by mathematical induction
 $P(n) : 3^n > n^3$ is true for all integers.

Q3 Use mathematical induction to show that for $n \geq 1$.

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

Solⁿ Let $P(n)$ be the given statement

$$\text{Base step: } P(n) : \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

$$\text{for } n=1, \frac{1}{1 \cdot 2} = \frac{1}{1+1} \Rightarrow \frac{1}{2} = \frac{1}{2}$$

$P(1)$ is true.

Inductive step: Assume that $P(k)$ is true

$$P(k) : \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1} \quad \text{--- (1)}$$

$$P(k+1) : \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

Adding $\frac{1}{(k+1)(k+2)}$ on both sides of $P(k)$ or eqⁿ (1).

$$\begin{aligned} \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2) + 1}{(k+1)(k+2)} \\ &= \frac{k^2 + 2k + 1}{(k+1)(k+2)} \end{aligned}$$

$$= \frac{(k+1)(k+1)}{(k+1)(k+2)}$$

$$= \frac{(k+1)}{(k+2)}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

Hence, by the principle of mathematical induction $P(n)$ is true for $n \geq 1$.